[WORKING TITLE — PI] The Matrix State Is Real: Minimal-Rank Recruitment, a Recall Capability Baselines Lack, and a Scale-Worsening Write-Geometry Pathology

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Abstract

Fast-weight and linear-attention models maintain a matrix-valued state, written by an outer-product update and evaluated mainly as an efficiency substitute for attention. We ask instead what the matrix state stores, at what dimensionality, and what the storage costs at scale: three pre-registered findings. First, a rank law: in a matrix-state encoder family trained on five permutation-group word problems (minimal faithful representation dimension 2 to 5), recruited effective rank tracks $d_{\min}$ (Spearman $\rho = 0.9747$, the tie-capped maximum), a designed S_4-versus-A_5 test shows rank follows dimension, not solvability, and forcing rank one below $d_{\min}$ zeroes recovery while $d_{\min}$ restores it. Second, a capability separation: a two-layer delta-rule model performs episodic recall at accuracy 0.9995 against a parameter-matched vector-state ablation at chance, the pre-registered decisive comparison; a compute-matched transformer also reads chance and is disclosed as a degenerate baseline. The capability resides causally in the first layer's state, is linearly legible only after downstream nonlinear processing, and holds at 0.998 or higher to 1798 tokens on a fixed 32,768-byte state while that transformer, cache-capped, reads chance everywhere. Third, the same write mechanism drives a population-geometry pathology that worsens monotonically from 14M to 1.31B parameters. Removing qk-normalization leaves it unchanged, and a frozen-key-bias mitigation fails to remove it: only the mitigation's loss neutrality survives the move to scale.

1 Introduction

Linear-attention and fast-weight sequence models replace the growing key-value cache of a transformer with a fixed-size matrix state, written by an outer-product rule and read by a matrix-vector product. The delta-rule family (DeltaNet and its gated descendants; Yang et al., 2025b;a) is the current standard bearer: at each step the state is updated as $S_t = S_{t-1}(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$, a rank-one correction that overwrites the value previously associated with the incoming key. The literature evaluates these models almost exclusively on one axis: quality per unit of compute or memory relative to attention. The state itself is treated as an implementation detail.

We take the opposite view: the $d \times d$ state itself is the object of study, a representational medium with a capacity structure and a causal role of its own, and with failure modes that never appear on a perplexity leaderboard. This paper measures all three, under pre-registered verdict criteria, across three experimental programs that share one object of study: what the matrix state stores.

Our thesis, scoped per leg to the architecture family that carries it: matrix-valued state representations are a genuine representational medium. In a matrix-state encoder family trained on group word problems, stochastic gradient descent recruits exactly the task-minimal state rank, and forcing one rank fewer destroys the task while restoring that rank restores it. In a two-layer delta-rule model, the first layer's matrix state causally carries an episodic-recall capability that a parameter-matched vector-state ablation lacks at equal training budget (a compute-matched transformer also fails, disclosed as a degenerate baseline), and the stored content is linearly legible only after downstream nonlinear processing. The same write mechanism, run at language-model scale, drives a population-geometry pathology that worsens monotonically across a two-decade parameter ladder and survives the community's stock mitigation.

Concretely, we contribute:

1. A rank law, causally closed (Section 3). Across five permutation groups whose minimal faithful representation dimensions $d_{\min}$ span 2 to 5, recruited effective rank tracks $d_{\min}$ at Spearman $\rho = 0.9747$, the maximum the family's tie structure permits, with 19 of 19 unconstrained cells inside the pre-registered band. A designed pair, S_4 versus A_5, shares $d_{\min} = 3$ but differs in solvability; an equivalence test declares their recruited ranks equivalent; solvability leaves no signature. Train-time rank forcing closes the causal loop: recovery is 0.000 at $k = d_{\min} - 1$ in all five groups and in all four seeds of the extension group, and returns at $k = d_{\min}$ in five of five. 2. A capability separation at matched budget (Section 4). On single-pass episodic recall, the delta-rule contender reads accuracy 0.99951 against 0.03394 for its vector-state ablation; the paired confidence interval excludes the pre-registered 0.30 margin, its floor sitting at more than three times the margin, and this ablation comparison carries the verdict. A compute-matched transformer also reads chance (0.02832); because its learning rate was never searched on this task and its training loss is near flat, it is recorded as a degenerate-baseline datum, not a second verdict (Sections 4.1 and 6). Zeroing the first layer's state collapses recall to chance while zeroing the second layer's changes nothing, and no state-level linear probe reads the content that the model's own forward pass decodes. The capability is also memory-stable: accuracy 0.998 or higher out to 1798 tokens on a fixed 32,768-byte state, while a transformer whose cache is capped at 1 to 32 times that budget reads chance everywhere. 3. A pathology that scale makes worse and the stock fix does not remove (Section 5). The write keys of delta-rule language models drift toward a collapsed population geometry; the span fraction of that drift climbs 0.248 to 0.455 across a 14M-to-1.31B ladder at held-fixed data mixes. Removing qk normalization changes nothing (0.05σ at n=3), so the attractor is not an artifact of the standard stabilizer, and a frozen-key-bias mitigation that is loss-neutral at every scale fails to transfer its 14M geometric benefit to 98M or 392M. 4. A mechanism scaffold (Appendix A). In the encoder family the trained state decomposes as $Z \approx c^* I +$ a rank-$(K-1)$ task correction at 0.5 to 2.9 percent residual, and deviation from this scaffold is a loss-blind health signal; the corresponding channel is empty by construction in delta-rule states, which bounds how far the scaffold generalizes.

Every numerical claim in this paper traces to a pre-registered verdict record and an archived, checksummed raw artifact; the figures regenerate from those artifacts under checksum assertion. Each repaired instrument reading appears together with the defect it corrected; Section 2 summarizes this discipline and Section 7 discusses its scope limits.

2 Background and Experimental Setup

2.1 Delta-Rule Fast-Weight States

A delta-rule layer maintains a matrix state $S \in \mathbb{R}^{d \times d}$ updated per token as $S_t = S_{t-1}(I - \beta_t k_t k_t^{\top}) + \beta_t v_t k_t^{\top}$, with keys k_t, values v_t, and write strength β_t produced from the token stream; reads are matrix-vector products $S_t q_t$. The update first erases the value currently bound to k_t, then writes v_t at k_t: an associative store with in-context writes. Because outer-product writes confine S to the span of the observed keys and values, the state's effective dimensionality and the geometry of the key population are both measurable objects, and both are studied here.

2.2 Model Families

Three model families appear, one per result leg. (i) The *matrix-state encoder* family (Section 3): a transformer encoder that emits a single $d_{state} \times d_{state}$ matrix Z per input word under a hard single-state bottleneck; the decoder reads only Z, never the raw inputs, so the task load must reside within the one matrix. (ii) The *two-layer delta-rule contender*

(Section 4): a DeltaNet-family language model with $d_{model} = 256$, two blocks, $d_{state} = 64$, and its two matched baselines, described in Section 4.1. (iii) *Delta-rule language models* at scale (Section 5): the same architecture family trained as causal LMs at 14M, 98M, 392M, and 1.31B parameters on held-fixed data mixes.

2.3 Tasks

Group word problems (Section 3). A word $w = g_{i_1} \cdots g_{i_L}$ is drawn over a symmetric generating set of a permutation group G; the target is the image of the composed element under a fixed faithful orthogonal representation. Five groups are used: S_3, S_4, A_5, S_5, A_6, with minimal faithful real representation dimensions $d_{\min} = 2, 3, 3, 4, 5$. The pair S_4/A_5 is the designed dissociation control: equal $d_{\min}$, opposite solvability. Train words have length $L \leq 8$; held-out evaluation words are fresh draws.

Episodic recall (Section 4). Each episode binds K=32 key-entity/value-entity pairs (real GPT-2 token identities, injectivity-guarded) over a 224-token bind phase, then queries one key. The metric of record, acc_A, is episode-restricted 32-way top-1 accuracy at the answer position, read through each model's own LM-head route; chance is $1/32 = 0.03125$ and the pre-registered demonstration bar is three times chance, 0.09375. Every acc_A number in this paper carries the same caveat: recall here means episode-restricted top-1 retrieval under argmax decoding, and under argmax a rank-one state can support on the order of d associations (Nichani et al., 2024); no rank or continuous-capacity claim is made from acc_A.

Language modeling at scale (Section 5). Causal LMs are trained on two fixed data mixes (a reasoning mix built on OpenR1-Math and a narrative mix built on WikiText-103, each with a fixed augmentation source), held identical across the parameter ladder so that scale is the only moving axis.

2.4 Instruments

Restricted effective rank (Section 3). States $Z(w)$ for held-out words are centered, the dominant $d_{\min}$-dimensional subspace U is taken from the SVD of the centered second-moment matrix, and effective rank (the entropy-based Roy-Vetterli measure) is computed on $U^\top Z(w) U$; fitting and evaluation use disjoint word splits. Centering is load-bearing: without it, an ambient identity block masquerades as signal.

Force-rank arms (Section 3). At train time, Z is projected to rank k by SVD truncation inside the forward pass, for $k \in \{d_{\min}-1, d_{\min}, d_{\min}+1\}$; recovery is measured as the fraction of held-out words whose Procrustes-aligned cosine to the target exceeds 0.9 (xrec90).

State zeroing and probe taps (Section 4). Causal attribution zeroes one layer's cached state before the query pass and re-measures acc_A; representational legibility is measured by ridge probes at four taps (three state-level, one post-block pre-LM-head), reported as the recovered fraction at cosine 0.9 (rf@0.9) against a shuffled control.

Span fraction (Section 5). For a population of K normalized write keys E, the raw instrument is the key-Gram deviation $\|E^\top E - I\|_F$; span fraction rescales it between two analytic anchors, the expectation for K i.i.d. random unit vectors (value 0) and full collapse onto one direction (value 1), making readings comparable across state widths. Higher is closer to collapse.

Confidence intervals. Paired confidence intervals in this paper are Student-t intervals on n=3 per-seed paired differences (two degrees of freedom, $t_{.975} = 4.303$), except the nine-seed diagnosis pool of Section 4.5, stated there. Per-seed values are tabulated in Appendix B so any alternative construction can be applied; the verdicts that rest on these intervals clear their margins by multiples of the interval width, and the one threshold-adjacent reading claimed as a trend (Section 5.2) is reported as unconfirmed.

2.5 Instrument Validity as a First-Class Method

Each leg of this program survived at least one instrument-defect-found-and-fixed round; only repaired readings appear below. The rank program's first causal sweep was voided by a target-padding defect that let rank-capped models buy cosine from an ambient identity block; the diagnosis, fix, and re-run are described in Section 3.3. The recall program lost three consecutive probe rounds to a wrong-layer, wrong-route linear instrument before the repaired tap of Section 4.3 read what the model's own forward pass had decoded all along. A third case came from a separate composition program (not claimed here): a primary lens read zero on converged models while a pre-registered crosscheck lens read one, the contradiction was resolved by a recorded tiebreak against the raw artifacts, and the affected endpoint is excluded from this paper entirely. Every verdict-bearing instrument here therefore carries an analytic anchor or a negative control, with a pre-registered crosscheck wherever a verdict hangs on a single metric.

3 The Rank Law

In the encoder family, the task's representation theory sets the state's dimensionality.

3.1 Recruited Rank Tracks Minimal Faithful Dimension

Figure 1 shows recruited restricted effective rank against $d_{\min}$ for the five groups, per seed. The per-group means are 1.877 (S_3), 2.852 (S_4), 2.832 (A_5), 3.591 (S_5), and 4.736 (A_6) against $d_{\min} = 2, 3, 3, 4, 5$. Spearman correlation across the family is $\rho = 0.9747$. This number is the maximum the family permits: the S_4/A_5 tie at $d_{\min} = 3$ caps the achievable midrank-adjusted ρ at exactly 0.9747, and the observed ordering attains that cap. Because the family has only five members, we report the exact permutation null rather than an asymptotic one: under the null, $P(\rho \geq 0.8) = 8/120 \approx 6.67\%$. The correlation is therefore suggestive on its own; the band criterion and the causal test below do the deciding. The tracking also holds cell by cell: all 19 unconstrained-arm cells in the sweep land inside the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band.

3.2 The Designed Dissociation: S_4 versus A_5

S_4 is solvable; A_5 is the smallest non-solvable group. Both have $d_{\min} = 3$. If recruited rank were driven by computational difficulty in the Krohn-Rhodes sense, the pair should separate; if by representation dimension, the pair should land together. We pre-registered a two-one-sided-tests equivalence procedure (Welch, $n = 5$ per group, margin ± 0.5 rank-units, half the ladder's adjacent-rung spacing). The result is a declaration of equivalence: mean difference 0.0194 rank-units (2.852 versus 2.832), standard error 0.0368, $df = 7.83$, with both one-sided statistics ($t_1 = 13.06$, $t_2 = 14.12$) exceeding the critical value 1.865 by a factor of seven. Recruited rank tracks the representation dimension of the task; the group's algebraic complexity class contributes nothing detectable.

3.3 An Instrument Defect, Its Diagnosis, and the Repaired Causal Test

An early force-rank sweep produced a mechanically valid FALSIFY reading that a target-embedding defect later voided, and the defect changes what the causal test means. The training target had been embedded as $\rho(g) \oplus I_{d_{state} - d_{\min}}$: an eye-padded block matrix of full rank d_{state} with all singular values equal to one. For such a target, a rank-k model's optimal direct cosine is $\sqrt{k/d_{state}}$ independent of which subspace it spends its rank on, and the harvest matched this prediction: 36 of 39 force-rank cells sat within 0.07 of the predicted ceiling, one further cell sat marginally outside at 0.072, and the two remaining outliers (0.15 and 0.17, both in one group's below-$d_{\min}$ arm) reflect an additional optimization shortfall. Rank-capped models were buying cosine from the identity padding, so the causal boundary at $d_{\min}$ was never tested. The fix replaced the padding with zeros, making the target rank exactly $d_{\min}$, and the decisional metric was pre-registered before the re-run as the

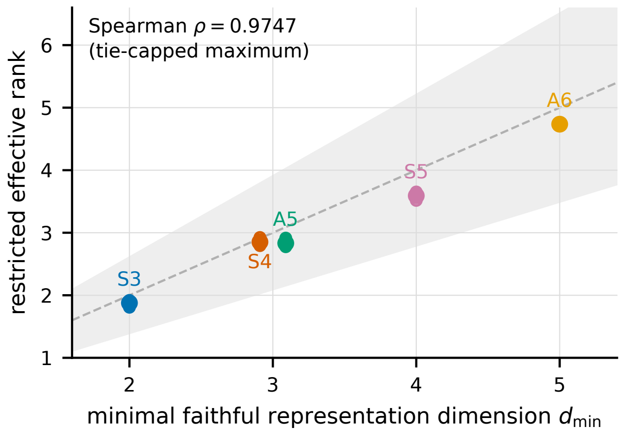

Figure 1: Recruited restricted effective rank versus minimal faithful representation dimension $d_{\min}$ for five permutation groups (S_3, S_4, A_5, S_5, A_6; unconstrained arms, per-seed points, group means marked). The shaded region is the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band; all 19 cells land inside it. Spearman $\rho = 0.9747$ equals the tie-capped maximum for this family. The S_4/A_5 pair (both $d_{\min} = 3$, opposite solvability) lands together; the equivalence test in Section 3.2 declares them equivalent within ± 0.5 rank-units.

Procrustes-aligned crosscheck (xrec90), because the scale-only primary metric reads cosine 0.01 to 0.22 on a flawless oracle when the target's rho-block eigenvalues are degenerate.

3.4 Recovery Is a Step Function at Exactly $d_{\min}$

Figure 2 shows the repaired force-rank staircase. At $k = d_{\min} - 1$, held-out recovery xrec90 is 0.000 in all five groups: not degraded, zero. At $k = d_{\min}$, recovery returns and clears the pre-registered bar of 0.9 times each group's own unconstrained anchor in four of five groups directly (S_4 0.800 against bar 0.585; A_5 0.700 against 0.630; S_5 0.600 against 0.450; A_6 0.650 against 0.585). S_3 landed within the pre-stated marginality window (0.450 against bar 0.495) and was routed to its pre-registered seed extension: across four independent seeds the seed-mean at $k = d_{\min}$ is 0.5625, clearing the bar, and $k = d_{\min} - 1$ reads exactly 0.000 in all four seeds. The pre-registered confirmation criterion (a razor step at $d_{\min}$ in at least four of five groups, including at least one of the marquee pair) is met with both marquee groups confirming.

Together the three readings close the loop in both directions: $d_{\min}$ is recruited when available and is causally necessary and sufficient when forced. The substrate is small and every reading reproduces from the archived artifacts (Appendix B).

4 Capability Separation

The head-to-head comparison reported here was designed, margined, and frozen before its confirmatory sweep ran: a two-layer delta-rule model against a parameter-matched vector-state ablation and a compute-matched transformer, on single-pass episodic recall.

4.1 Arms, Matching, and Frozen Margins

The contender is a two-block DeltaNet-family LM ($d_{model} = 256$, $d_{state} = 64$, 14,049,408 parameters). The ablation keeps the embedding table, output head, and feed-forward blocks identical and replaces only the fast-weight mixer with an elementwise-gated vector recurrence at the same d_{state} (14,048,384 parameters, a 0.007 percent difference); it is constructed as a recurrence rather than a reshape, which makes matrix structure the manip-

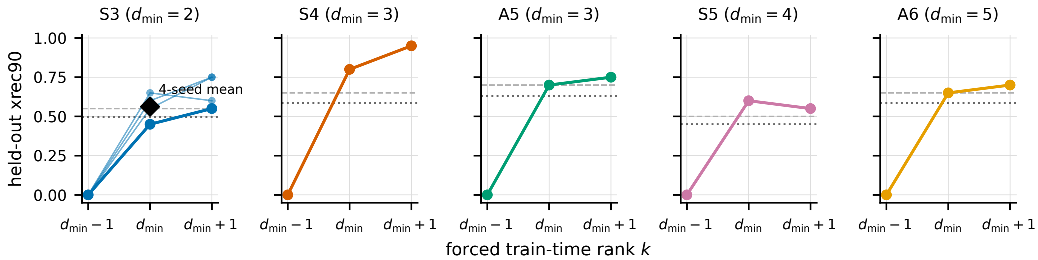

Figure 2: Held-out recovery (xrec90, Procrustes-aligned cosine threshold 0.9) versus forced train-time rank k, per group, on the zero-padded (repaired) target. Recovery is 0.000 at $k = d_{\min} - 1$ in every group and returns at $k = d_{\min}$; dashed lines mark each group's bar of 0.9 times its own unconstrained anchor. The S_3 panel overlays all four extension seeds: every seed reads 0.000 at $k = d_{\min} - 1$, and the seed-mean at $k = d_{\min}$ (0.5625) clears the bar (0.495).

Table 1: Episodic recall acc_A per seed and arm at matched parameters, tokens, and compute; chance $1/32 = 0.03125$, demonstration bar 0.09375.

arm	seed 0	seed 1	seed 2	mean
contender	0.99951	1.00000	0.99902	0.99951
vector-state ablation	0.03223	0.03271	0.03687	0.03394
transformer	0.02710	0.02930	0.02856	0.02832

ulated variable at near-identical parameter count. The transformer baseline is a two-layer pre-norm decoder with rotary position embeddings and the contender's own MLP class, FLOP-matched to the contender within 5 percent. On this task its learning rate was the shared default 3×10^{-4}, identical to the contender's and the ablation's; the calibration record's three-point learning-rate grid was run on the language-modeling control task, and no architecture-specific learning-rate search was performed for the transformer on the recall task. Its recall-task training loss is near flat across the run (7.81 to 7.84 at step 500, 7.45 to 7.51 at step 20,000, across the three seeds), a signature consistent with an under-optimized arm; Sections 6 and 7 state the resulting interpretive limit and the measurement that would resolve it. All arms train 20,000 steps at matched tokens with seeds identical across arms per cell. Verdict tiers, margins, and the primary task were frozen and checksummed before the confirmatory sweep launched (a WIN requires the paired per-seed difference's 95 percent confidence interval to exclude 0.30); the sweep's evaluation applied the audited round protocol verbatim to its 18 checkpoints with checksum-pinned loading and a fixed evaluation episode set.

4.2 The Verdict at n=3

Table 1 (Figure 3, left) gives the per-seed metric of record.

Every contender seed exceeds the demonstration bar by a factor of at least 10.7; neither baseline clears it in any seed. The paired contender-minus-ablation difference is 0.96558 with 95 percent confidence interval (0.95822, 0.97293), and the contender-minus-transformer difference is 0.97119 with interval (0.96855, 0.97383); both intervals exclude the frozen 0.30 margin, the first by more than three times the margin at its floor. The pre-registered extension trigger (any seed below bar, or any interval straddling the margin) did not fire.

The Nichani caveat of Section 2.3 is part of the claim: recall means episode-restricted top-1 retrieval under argmax decoding, under which a rank-one state can support on the order of d associations, so nothing here is a rank or continuous-capacity claim. So is the matched-budget caveat: the transformer read holds only at this matched parameter count, token budget, and training compute. Because the transformer sits below its own demonstration bar, it is recorded as a degenerate-baseline datum, and the separation verdict is carried by the ablation comparison with the transformer's non-competitiveness disclosed alongside.

4.3 The Capability Is Fast-Weight-Resident and Nonlinearly Stored

Causal attribution by state zeroing localizes the capability. On the frozen round checkpoint, zeroing the first block's state before the query collapses contender recall from 0.9990 to 0.0286, at chance, while zeroing the second block's state leaves it at 0.9990, unchanged. The sweep replicates this on all fresh seeds: first-state zeroing reads 0.0339, 0.0012, and 0.0002 against the 0.09375 bar, and the hard-stop criterion is clean in 12 of 12 recurrent cells. The binding lives in S_0; S_1 is causally inert for this task.

Where the content becomes readable is a separate question. Ridge probes at three state-level taps, including directly on the causally load-bearing S_0, recover nothing at the strict threshold (rf@0.9 = 0.0 at every state-level tap; the accompanying shuffled-control gaps of at most 0.063 are mean-cosine gaps, a separate continuous sanity check on a different scale from rf@0.9). The same probe at the post-block-1, pre-LM-head hidden state reads rf@0.9 = 0.674 with mean cosine 0.894. The ablation's pre-LM-head tap, its own positive control, fails (rf@0.9 = 0.0, cosine 0.119); its pre-LM-head geometry is different in kind. The stored binding is therefore real but linearly illegible at the state; it becomes linearly decodable only after the downstream block's nonlinear processing. A state-level linear probe alone would have read this capability as absent, while the model's own forward pass decodes it at 0.9995.

4.4 Constant-Memory Recall Horizon

The contender's two states occupy $2 \times 64 \times 64$ float32 entries, 32,768 bytes, independent of context length. Holding those bytes fixed, recall does not decay with distance: at horizons of 2, 4, and 8 times the bind phase (454, 902, and 1798 tokens), every seed reads $\mathrm{acc}_A \geq 0.998$. The pre-registered memory-matched comparison caps the transformer's KV cache at M times the contender's state bytes with sink-plus-FIFO eviction in the style of streaming attention caches (Xiao et al., 2024), $M \in \{1, 2, 4, 8, 16, 32\}$; every capped read at the decision horizon lies between 0.020 and 0.033, at or below chance, as does the uncapped read. Every per-M paired-gap confidence interval has a floor of at least 0.958 against the 0.20 crossover margin. Because the uncapped transformer itself fails the demonstration bar, the pre-registered degenerate-baseline clause fires: the verdict of record is that the baseline is non-competitive at matched parameters and tokens, not a strongest-possible-baseline result, and no crossover-point value is certified. The two informative readings are the contender's own flat horizon table and the answer to the forced-locality question: capping never helps the transformer.

4.5 Scope Disclosure: the Multi-Hop Task

A second task in the frozen registry, two-hop compositional recall, is verdict-INDETERMINATE and is not claimed. Its data are disclosed because they constrain interpretation: one of three contender seeds clears the demonstration bar (0.33447; the other two and all baseline seeds read chance), the paired interval straddles the margin, the bar-clearing seed's partial recall does not generalize to held-out hop depths (0.0112) and collapses to 0.010 at every extended horizon. The single bar-clearing seed suggested the failure mode is at least partly trainability variance rather than a hard capability boundary, and the pre-registered diagnosis round confirmed this: across nine pooled seeds, three contender seeds clear the demonstration bar (0.334, 0.391, 0.479) while zero of nine ablation seeds do; the hard-capability-boundary hypothesis is rejected for this task at this scale and budget. The pooled paired interval $(-0.020, +0.268)$ sits below the 0.30 margin and the strict-tier reading is a TIE, flagged non-decision-grade by a pre-registered batch-effect gate; every bar-clearing seed remains fragile at held-out hop depths and extended horizons. Nothing beyond single-hop recall is claimed here.

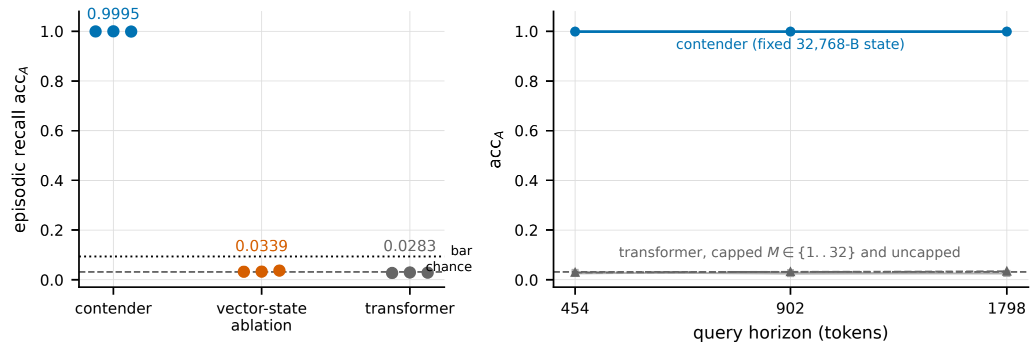

Figure 3: Left: episodic recall acc$_A$ per seed and arm at matched parameters, tokens, and compute (K=32 bindings; chance 0.03125, dashed; demonstration bar 0.09375, dotted). The contender reads 0.99902 to 1.00000; both baselines sit at chance in every seed; paired intervals exclude the frozen 0.30 margin. Right: recall versus context horizon on a fixed 32,768-byte contender state (454 to 1798 tokens, all seeds at or above 0.998) against the KV-capped transformer grid ($M \in \{1..32\}$ times the contender's state bytes, all reads 0.020 to 0.033). Recall means episode-restricted argmax retrieval (Nichani caveat, Section 2.3); the transformer is a degenerate-baseline datum at this matched budget.

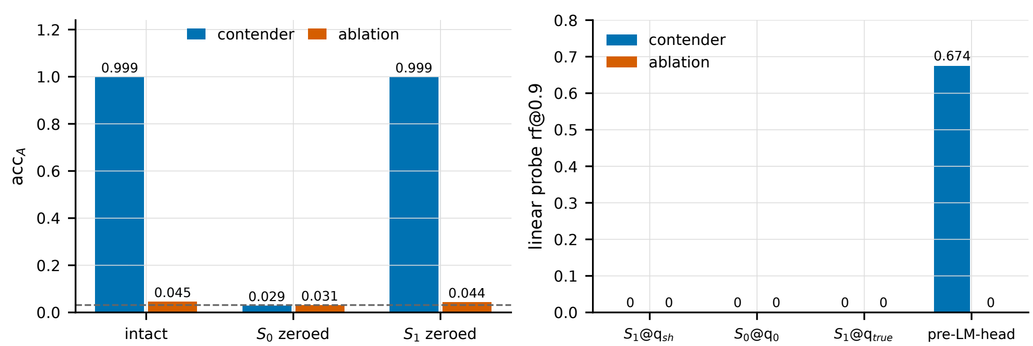

Figure 4: Storage localization and legibility for the contender and its vector-state ablation. Left: acc$_A$ with both states intact, first state zeroed, and second state zeroed; first-state zeroing collapses the contender to chance (0.9990 to 0.0286) while second-state zeroing changes nothing. Right: linear-probe recovered fraction at cosine 0.9 for three state-level taps and the pre-LM-head tap; only the contender's pre-LM-head tap reads above zero (0.674), so the binding stored in S_0 is linearly legible only after downstream nonlinear processing.

5 The Pathology at Scale

The same write mechanism that Sections 3 and 4 show storing task structure also reshapes the population geometry of its own keys, for the worse, and more so at larger scale. The evidence here is independent of Sections 3 and 4.

5.1 The Write-Geometry Attractor Worsens Monotonically with Scale

In trained delta-rule LMs, the write-key population drifts away from the near-orthonormal geometry of random unit vectors toward a collapsed, low-dimensional configuration. Span fraction (Section 2.4) locates each model between the random anchor (0) and full collapse (1). Across a four-rung ladder trained on held-fixed data mixes, at 14M, 98M, 392M, and 1.31B parameters, pooled span fraction reads 0.248, 0.344, 0.389, and 0.455: monotone in scale, with the largest model halfway to collapse. Because the data mixes are held fixed across rungs, scale is the only moving axis.

The 1.31B cells self-terminated at their internal timeout at approximately 84.7 percent of the token-matched budget. The shortfall does not rescue the trend: over the final 25,000 recorded steps the rung's span fraction is flat to mildly declining (0.4584 to 0.4554), so completing the budget would move the endpoint by roughly 0.003, more than twenty times smaller than the 0.066 increment from the previous rung.

5.2 Removing qk Normalization Changes Nothing; Gating Trends Toward Amplification

The entire ladder was trained with qk normalization active, the same in-kernel key normalization that production linear-attention stacks cite as their stability mitigation. A 2-by-2 factorial at 14M (qk-norm on/off crossed with delta-rule gating on/off, n=3 seeds, pre-registered same-corpus noise floor 2.244355 raw Gram-deviation units) gives both answers. Removing qk normalization changes the geometry reading by -0.103, which is 0.05σ against the floor: a within-noise null. The attractor is measured with the mitigation active and is unchanged without it; it is not an artifact of the stabilizer. Turning on gating reads $+4.312$, which is 1.92σ, below the pre-registered 2σ confirmation bar of 4.489; all three paired seeds move in the amplifying direction and an exploratory Welch test gives $p = 0.062$. We record gating as a direction-consistent trend toward amplification: not a confirmed effect, and not a null.

5.3 The Small-Scale Fix Does Not Transfer

At 14M, a frozen-key-bias construction had moved the geometry: blending each key with a frozen per-token random unit anchor, $k \leftarrow \text{normalize}((1-\lambda)k + \lambda B[\text{tok}])$ at $\lambda = 0.58$, with zero new trainable parameters. A pre-registered transfer wave tested it at 98M and 392M against an eval-retrofitted comparator (the baseline checkpoint measured through the identical blend, which removes the mechanical pinning artifact a naive comparison introduces), on both corpora, with a pre-blend co-primary instrument that measures the raw keys and is immune to the blend arithmetic by construction. The outcome is the pre-registered PARTIAL verdict at both scales. At 98M the deployed arm is destabilizing, the 14M sign, on both corpora and both instruments: span-fraction deltas $+0.1133$ with interval $(+0.0543, +0.1723)$ on the reasoning mix and $+0.1011$ with interval $(+0.0541, +0.1482)$ on the narrative mix. At 392M the narrative-mix delta is $+0.0189$ with interval $(+0.0112, +0.0266)$, still destabilizing, while the reasoning-mix interval includes zero. The effect reverses nowhere. A single-seed exploratory probe of the alternative global-vector construction decays from -0.33 and -0.23 at 14M to -0.058 and -0.034 at 98M, and at 392M reads -0.012 on one corpus and $+0.019$, a sign flip, on the other: the stabilization it bought at small scale decays toward zero and loses its sign. Meanwhile the constructions are free at the loss level: validation-loss neutrality passes its blind-pinned ceiling in all eight gate cells under the pre-registered arm-mean criterion; one individual seed in the 98M narrative-mix cell reads 3.2038 against the 3.2020 ceiling while its arm mean (3.1961) passes. The neutrality transfers to scale. The fix does not. The 392M token budget bounds the cross-scale comparison: those cells ran a reduced budget, so the apparent 98M-to-392M attenuation is confounded with tokens by design and is not a registered claim; the within-scale readings are the registered claims.

5.4 The Storage Mechanism as Common Factor

The write that Sections 3 and 4 show storing task structure is the write that, iterated over a corpus at scale, drags its own key population toward collapse; the one construction known to stabilize the geometry at 14M buys nothing at 98M and 392M while costing nothing in loss. No fix is offered here: the community's stock mitigation is orthogonal to the phenomenon, and the obvious portable construction fails to port. The pathology belongs to the storage mechanism as scaled; no stabilizer choice tested here explains it away, and it is currently unaddressed.

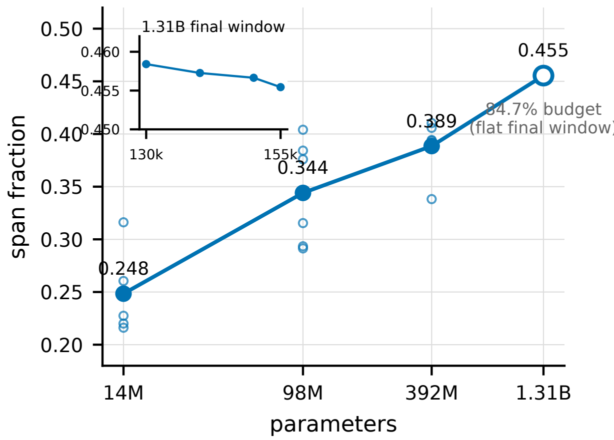

Figure 5: Write-key span fraction versus model scale (14M, 98M, 392M, 1.31B parameters; log-scale abscissa) on held-fixed data mixes; 0 is the random-unit-vector anchor and 1 is full collapse, computed per state width so rungs are comparable. The climb 0.248 to 0.455 is monotone. The 1.31B rung stopped at approximately 84.7 percent of its token budget with span fraction flat to declining over its final window (0.4584 to 0.4554); the missing budget cannot supply the 0.066-scale increment.

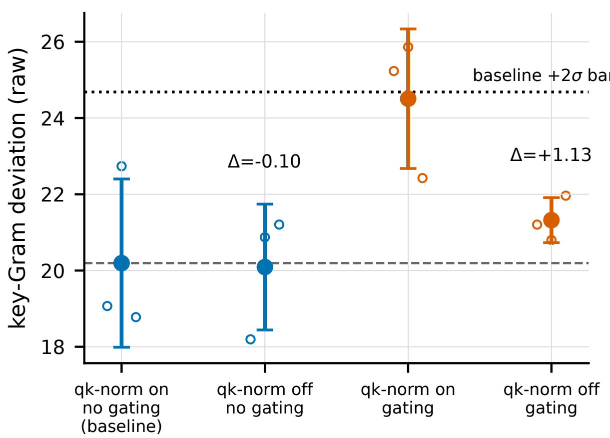

Figure 6: The 2-by-2 mitigation factorial at 14M (n=3; raw key-Gram deviation, cell means with per-seed points; dashed line: the pre-registered 2σ bar over the corrected same-corpus noise floor). Removing qk normalization is a within-noise null (-0.103, 0.05σ): the attractor is not a qk-normalization artifact. Gating reads $+4.312$ (1.92σ, all three paired seeds positive), a direction-consistent trend below the confirmation bar: neither a confirmed amplification nor a null.

6 Related Work

Associative capacity of fast-weight states. Nichani et al. (2024) prove that under argmax decoding a rank-one matrix can recover on the order of d associations, which is why no result in this paper derives a rank or capacity claim from argmax recall: the rank law of Section 3 forces exact continuous recovery through a Procrustes-aligned cosine readout, and every argmax recall number in Section 4 carries their caveat explicitly. Their analysis is the reason the causal razor of Section 3 refuses argmax decoding anywhere a rank claim depends on it.

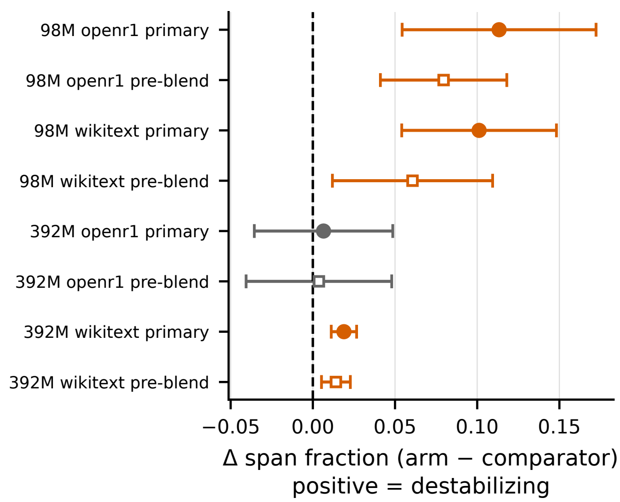

Figure 7: Transfer of the frozen-key-bias mitigation to scale: per-scale, per-corpus span-fraction deltas (deployed arm minus eval-retrofitted comparator, n=3, 95 percent intervals; negative would be stabilizing). At 98M both corpora read destabilizing with intervals excluding zero; at 392M one corpus reads destabilizing and one is null; no cell reverses. Validation-loss neutrality passes in all eight gate cells (not shown), so the construction's cost neutrality transfers while its geometric benefit does not. The 392M rung ran a reduced token budget; cross-scale attenuation is therefore not a registered claim.

Associative recall in transformers. Multi-query associative recall is a regime where softmax attention is documented as strong: Arora et al. (2023) introduce the MQAR benchmark and show attention solves it where sub-quadratic models struggle, Jelassi et al. (2024) report the same advantage for copying over state-space models, and even two-layer from-scratch transformers reliably develop induction circuitry (Olsson et al., 2022). Our transformer baseline's chance-level reading therefore runs against the expectation this literature sets, and we do not interpret it as an architectural inability. The leading candidate explanation is optimization: the arm trained at the shared default learning rate with no architecture-specific search on the recall task, and its training loss is near flat across the run (Section 4.1). The separation verdict accordingly rests on the vector-state ablation comparison alone, with the transformer reading disclosed as a degenerate-baseline datum; Section 7 states the measurement that would resolve the question.

Rank measurement in linear attention. Two contemporaneous studies (Nazari & Rusch, 2026; Sun et al., 2026) measure the rank of linear-attention states descriptively, on frozen checkpoints. Section 3 differs in both directions of inference: the force-rank arms intervene at train time, and the necessity/sufficiency step at $d_{\min}$ is a causal result a descriptive measurement cannot license.

Key normalization and stability. The qk-normalization line (Kimi Linear, Team et al., 2025; Qwen3-Next) conditions single-vector eigenvalue stability of the state update. The attractor of Section 5 is a different object, cross-key population geometry. It was measured with qk normalization active throughout, and removing the normalization is a within-noise null at n=3.

DeltaNet and gated variants. DeltaNet (Yang et al., 2025b) and Gated DeltaNet (Yang et al., 2025a) supply the architecture substrate for the capability and pathology legs (Sections 4 and 5) and benchmark it for quality and throughput. This paper asks what the states represent: the rank law of Section 3 is established on a separate matrix-state encoder family with no delta-rule write, while the causal recall capability and the write-induced geometry at scale are established on the delta-rule substrate itself. The gating axis reappears here only as a measured factor in Section 5.2, where it trends toward amplifying the attractor without confirming.

Prior negative result in this program. A published companion study ("The Gradient Does Not See Rank," ICML 2026 MI workshop) showed that bolting a matrix state onto a pre-trained backbone through a flatten-then-project readout leaves the gradient rank-blind. The present paper is the from-scratch counterpart: when the state is native and the readout preserves structure, SGD does see rank (Section 3), and the resulting medium supports a capability its matched ablations lack (Section 4).

7 Discussion and Limitations

Scale scope of the capability legs. The rank law and the recall separation are small-model, synthetic-task results by design: the encoder family trains in minutes and the head-to-head contender is a 14M-parameter model on a controlled binding task. The strength of that regime is causal closure (train-time rank forcing, state zeroing, frozen margins); its limit is external validity. Nothing here shows the recall separation surviving at language-model scale or on natural data, and the matched-budget caveat bounds every transformer comparison: a differently scaled or differently trained transformer requires its own matched re-run before any extrapolation. The specific unresolved measurement is a learning-rate search for the transformer arm on the recall task itself: the frozen protocol's three-point grid was run on the language-modeling control task only, so the chance-level recall reading has not been separated from an optimization failure. The resolving experiment is a re-run of the transformer arm under the identical frozen protocol at a grid of at least four learning rates spanning 10^{-4} to 3×10^{-3}, three seeds, 20,000 matched steps, with training curves reported. A tuned transformer that still reads below the demonstration bar would strengthen the separation to a two-baseline result; a tuned transformer that clears it would confine the separation to the vector-state comparison, which is the comparison the frozen registration designates as decisive in either case.

One stress point is a locate-only null. At the disclosed above-capacity stress load ($K/d = 0.75$), the completed three-arm table reads chance in every arm: contender 0.0189, ablation 0.0195, transformer 0.0218, against chance 0.0208. The table is locate-only by pre-registration, not a verdict-grade cell; no capability claim is made at this load, and it bounds the single-hop recall separation to loads below this stress point at this training budget.

Scope of the pathology leg. The ladder result is a geometry claim: span fraction worsens monotonically with scale, but this paper does not tie the drift to a downstream performance cost, and validation loss is neutral in every mitigation cell measured. The tie between the geometric drift and any behavioral cost is the outstanding scientific question, and the 392M token-budget confound (Section 5.3) caps what the transfer wave can say across scales.

Multi-hop composition is open. The two-hop task is verdict-INDETERMINATE, with partial recall in three of nine pooled contender seeds, none generalizing to held-out hop depths (Section 4.5), and a separate compositional-depth program in this research line is at the design-and-calibration stage with a pre-registered lens question resolved but no model verdict; we claim neither. Depth generalization is future work, stated as such.

Instrument validity as a recurring risk. Three times in this program (Sections 2.5, 3.3, 4.3), the first instrument pointed at a true phenomenon read false. In each case the fix was minor once the defect was seen. Representational claims about matrix states need independent checks of the kind Section 2.5 catalogs, because errors in either direction arise from instrument choice alone at this scale: the wrong-tap probe read a real capability as absent, and the identity-padded target sold ambient structure as recovery.

8 Conclusion

A $d \times d$ state written by outer products is a measurable representational medium. Its trained rank is set causally by the task's representation theory in the encoder family. In the delta-rule family, the first layer's state causally carries recall that a matched vector-state ablation lacks (the compute-matched transformer, a degenerate baseline, also fails); the write keys of its language models drift toward collapse at every larger scale, stock normalizer or no. The storage and the drift come from one mechanism, and that is the design constraint this paper leaves behind: interventions aimed at the geometry must operate on the write rule itself rather than on its normalizers, with the Section 5 negative transfer result as the first measured bound on that search. Two measurements would extend the map from here: a recall-task learning-rate search for the transformer baseline, and a tie from the geometric drift to a behavioral cost.

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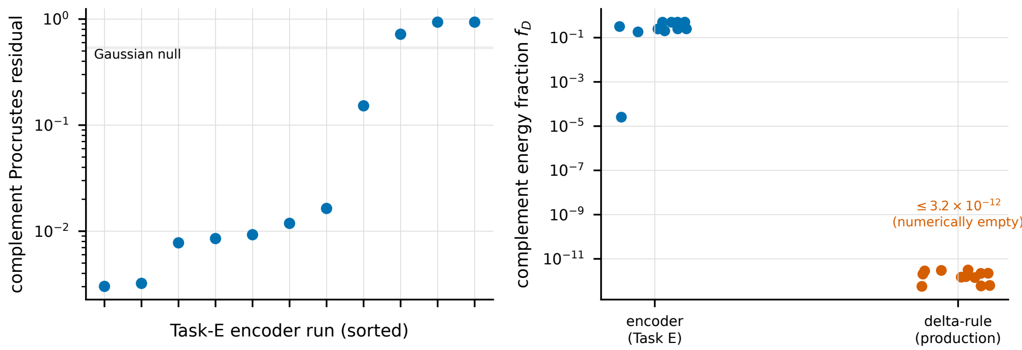

Figure A1: The complement scaffold is architecture- conditional. Left: complement Procrustes residual per archived Task-E encoder run (sorted; log scale). Converged runs sit at 0.003 to 0.018, two orders of magnitude below the Gaussian-null band (0.53 to 0.54, shaded); the high-residual runs are the archive's non-converged members, which is the separation behind the loss-blind health signal (Spearman $\rho = -0.973$ against held-out task cosine, all eleven runs). Right: complement energy fraction f_D by architecture family (log scale): the encoder family carries measurable complement energy, while delta-rule production states read at most 3.2×10^{-12}, numerically empty, because outer-product writes confine the state to the span of observed keys and values.

A The c^*I Complement Scaffold

A mechanism-level observation, architecture-conditional and offered as a candidate rather than a claim of the main thesis. In the converged matrix-state encoder family (the Section 3 substrate's sibling trained on parallel key-value binding), the trained state decomposes as $Z \approx c^* I_d + \Delta$, where Δ is a rank-$(K{-}1)$ task correction: the whole-state law holds at 0.5 to 2.9 percent Frobenius residual, with per-example identity alignment $\tau \geq 0.9997$. The effective rank of $Z - c^*I$ sits within 0.3 of the $K{-}1$ target at the $K{=}8$ load (deviations 0.03 to 0.26); at the $K{=}12$ load the three converged runs read 10.22 to 10.41 against the target of 11, deviations of 0.59 to 0.78. The rank-$(K{-}1)$ characterization is therefore exact-order at the lighter load and approximate at the heavier one. The scaffold is emergent: the architecture has no identity path or output gain that parameterizes it.

The scaffold earns its keep as a loss-blind health signal: the Procrustes residual of the complement correlates with held-out task cosine at Spearman $\rho = -0.973$ across all eleven archived runs, read entirely from a subspace the loss never constrains, so it can grade convergence without labels. Its boundary is architectural, because the scaffold does not exist where Sections 4 and 5 live: in delta-rule states the orthogonal complement channel is numerically empty (complement energy fraction at most 3.2×10^{-12} across 12 of 12 production-family runs), because outer-product writes confine the state to the span of observed keys and values by construction. The scaffold is therefore an encoder-family instrument and a delta-rule impossibility, which is itself a datum: the two families that share the rank law of Section 3 and the capability of Section 4 do not share their complement structure, and mechanism stories that depend on an identity scaffold cannot port across that boundary.

B Configurations, Full Tables, and Reproducibility

B.1 Model and Training Configurations

Setting	Value
Stage-1 encoder (Section 3)	h=32, 3 layers, $d_{state} = d_{\min} + 2$ per group; single-state bottleneck; step budgets 8K/20K/20K/8K/40K for $S_3/S_4/A_5/S_5/A_6$
Head-to-head arms (Section 4)	contender d_{model}=256, 2 blocks, d_{state}=64, 14,049,408 params; ablation 14,048,384 params; transformer depth 2, FLOP-matched $\leq$ 5%; all arms 20,000 steps, matched tokens, seeds identical across arms
Ladder rungs (Section 5)	14M/98M/392M/1.31B = $(d_{model}, L, d_{state})$ of (256, 2, 64) / (768, 12, 64) / (1536, 16, 128) / (2560, 22, 128); measured 14,048,896 / 97,618,176 / 391,869,440 / 1,311,135,488 params
Span-fraction anchors	random $\sqrt{K(K-1)/d}$, collapse $\sqrt{K(K-1)}$; at K=64: 7.94/63.50 (d=64), 5.61/63.50 (d=128)

B.2 Per-Seed and Per-Cell Tables

The camera-ready appendix carries the full per-seed tables regenerated from the archived artifacts named in Section B.3: the 19-cell unconstrained rank table with band edges (R1), the 39-cell first-sweep force-rank diagnostic with each cell's deviation from the $\sqrt{k/d_{state}}$ ceiling, the three cells outside 0.07 named (R1b), the 30-cell force-rank grid plus the 24-cell S_3 extension (R3), the 27-cell head-to-head sweep with all re-metric readings and hard-stop legs (R4), the localization and tap tables for both arms (R5), the per-rung ladder trajectories (R6), the 12-cell 2-by-2 factorial (R7), the 28-cell fix-at-scale wave with both instruments and all gate cells (R8), and the 90-pass memory-matched fan-out (R10). Every table is generated by the same script that generates the figures, under the checksum assertions of Section B.3.

B.3 Artifact Manifest and Figure Regeneration

Every figure in this paper is generated by a single versioned script (figures/figure-gen.py) that loads only archived result files and asserts the MD5 checksum of each file before plotting; a checksum mismatch fails the build rather than plotting stale data. The manifest below lists the verdict-bearing artifacts (checksums recorded 2026-07-09/10; the archive layout carries per-run manifests verified against the compute host).

Result	Artifact	MD5
R1/R2	2026-07-09__capability__sweep__harvest/harvest__summary.json	7dce77dcba724cd1004419ac71fe5f2f
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R3 (ext)	2026-07-09__m3fix__s3ext/harvest__analysis__output.txt	d4413385f9f45b71ff9707354ed6b055
R4	2026-07-10__h2h__sweep__harvest/ (27 training + 18 re-metric JSONs; margins token 58fe68e7b15728739a5176d7e591e204)	per-file manifest
R5	2026-07-09__h2h__tap__localization/results/tap__localization__{contender,ablation}.json	362333c8..., ff8e352a...
R6	2026-07-06__trackc__rung3/probe__analysis__rung3.json; 2026-07-06__trajectory__probes/trajectories__tidy.json	6a627c31..., 0fe53d8b...
R7	2026-07-09__attrrob__2x2__escalation__harvest/{box__results/AGGREGATE.json,n3__recompute__summary.json}	d7e4d6b4..., b4bdffdf...
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Full 32-character checksums for the abbreviated entries appear in the released manifest file alongside the figure script. The named (arXiv) build links the archive root; the anonymized build links an anonymized mirror.

B.4 Pre-Registration Records

Each verdict traces to a design-document record written before the confirmatory run: the rank-law sweep and its instrument repair (§1.33–§1.36a of the capability-separation design), the head-to-head margins freeze and sweep verdict (§1.31–§1.41 of the head-to-head design), the ladder and mitigation waves (the scale-transfer design and §13 of the frozen-bias design). The records fix scope, seeds, bars, and tiers; where a record and an artifact ever disagreed, the raw-artifact reading and the recorded tiebreak stand in place of the record's prose.